
TetriC

A CONSOLE-BASED TETRIS GAME

STUDENT

Md. Shourov (BSSE-1609)

SUPERVISOR

Toukir Ahammed

Lecturer, IIT, University of Dhaka

Project Goal & Status

Primary Goal

To develop a **robust console version** of the classic puzzle game Tetris, focusing on efficient logic and real-time user interaction within a terminal environment.

STATUS: Fully functional console game with split controls for simultaneous competition.

Evolution

Proposal Feedback: Initially focused on single-player logic. Based on feedback, the project evolved to include more complex interactions.

Midterm Update: Successfully integrated a **Local Two-Player Mode** on the same device, allowing two players to compete simultaneously.

Project Overview

Tetric is a console-based implementation of the classic Tetris game, designed for dual-player competition within a terminal environment.



Core Concept

A puzzle game where randomized shapes fall from the top. Players must arrange them to clear lines and score points.



Gameplay Loop

Continuous cycle of moving, rotating, and dropping shapes. Completed rows disappear, increasing the player's score.



Win Condition

The game ends when blocks reach the ceiling. In 2-player mode, the last surviving player is declared the winner.



Visual Structure

Features two independent 10x20 grids rendered side-by-side in the console for simultaneous local multiplayer.

Player Controls (Implemented)

PLAYER 1

- 1 Rotate
W Key
- 2 Move Left / Right
A / D Keys
- 3 Soft Drop
S Key
- 4 Hard Drop
Z Key

PLAYER 2

- 1 Rotate
Up Arrow
- 2 Move Left / Right
Left / Right Arrows
- 3 Soft Drop
Down Arrow
- 4 Hard Drop
Spacebar

Key Features (Completed)

Milestone 1

Core Gameplay Systems

The project implements all fundamental Tetris mechanics, ensuring a fair and responsive experience for both players in a shared console environment.

Technical Highlights

Random Generation: Fair distribution of all 7 Tetromino pieces (I, O, T, S, Z, J, L).

Split-Screen: Optimized buffer management for flicker-free simultaneous rendering.

Advanced Game Logic

- ✓ Independent line detection and clearing animations
- ✓ Precise collision detection for boundaries and blocks
- ✓ Real-time score updates for both players
- ✓ Automated Game Over detection and winner declaration

Feature	Implementation Status
Collision Logic	100% Verified
Line Clearing	Optimized
Dual Rendering	Flicker-Free

Tools Used

Tech Stack

Development Environment

The project was developed using industry-standard tools for C programming, ensuring code quality and version control.

 **Language: C Programming Language** - Chosen for its efficiency and low-level console control.

 **IDE: Visual Studio Code** - Used for its robust debugging and extension support.

 **Version Control: GitHub** - Managed project milestones and code history.



Core C Libraries

windows.h

Essential for cursor management, screen buffer control, and console coloring.

conio.h

Used for real-time keyboard input handling without blocking the game loop.

Technical Requirements

The project requires a **Windows environment** due to the use of specific system headers for console manipulation.

Compiler: GCC (MinGW) was used for building the executable.

Future Work & Roadmap

Phase 2

Networking & Gameplay

The next development phase focuses on transitioning from local competition to a networked multiplayer environment.

Key Enhancements

- Socket Programming: Client-Server architecture for remote play.
- Next Piece Preview: UI element for strategic planning.
- Dynamic Difficulty: Adaptive speed increase algorithms.
- Pause/Restart: Mid-match control functionality.

Current State	Future Vision
Local Multiplayer	Networked (Sockets)
Basic Console UI	Enhanced UI Elements
Fixed Speed	Dynamic Difficulty
Single Device	Cross-Device Play

Technical Improvements

Code Optimization: Modularizing redundant functions between Player 1 and Player 2 logic to reduce code footprint and improve maintainability.

Conclusion

The Tetric project has successfully reached its midterm milestone, demonstrating robust game logic and a functional dual-player experience.

✓ Project Achievements

Successfully met and exceeded initial project goals through iterative development.

Goal Fulfillment

Implemented core Tetris logic and expanded to local two-player mode.

Technical Proficiency

Demonstrated efficient use of C arrays, game loops, and system headers.

User Interaction

Created a responsive, flicker-free console interface for real-time play.

Key Takeaway:

"The project serves as a solid foundation for advanced C programming and game development concepts."

→ Next Steps

Transitioning towards a more advanced, networked version of the game.

Networking

Implementing Socket Programming for remote multiplayer capabilities.

UI Polish

Adding "Next Piece" previews and enhanced scoreboards.

Optimization

Refactoring code for better modularity and cross-platform support.

Vision:

"To create a fully networked, competitive console Tetris experience with modern UI features."

Tetric: A Console-Based Tetris Game

THANK YOU

